

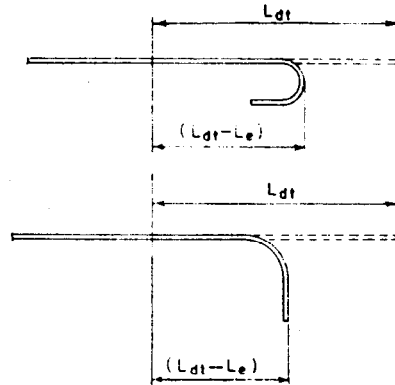
AMENDMENT NO. 1 MARCH 1989

TO

SP : 34 (S & T) - 1987 HANDBOOK ON CONCRETE REINFORCEMENT AND DETAILING

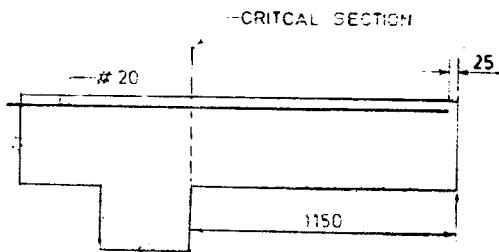
(Page 30, clause 4.3.1.2) — Add the following matter at the end of the clause:

If for one bar size, straight anchorage length, L_{dt} will develop the tensile design yield stress in tension, then by bending a standard hook or bend at the end, a length $(L_{dt} - L_e)$ will develop the design yield stress also. This aspect is illustrated in the figure given below, where L_{dt} is the development length in tension and L_e is the anchorage value of hook/bend. In some cases the length $(L_{dt} - L_e)$ will have a negative value, in which case it shall be assumed that the hook/bend alone provides an adequate development length.

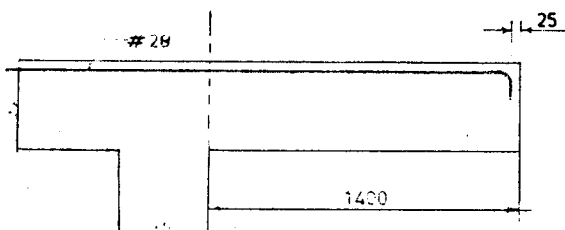


When hooks/bends do not conform to standard bends/hooks given in Table 4.1, anchorage value of hook/bend shall be neglected and the total development length provided (measured along bend/hook) shall be equal to the required development length (L_{dt}) .

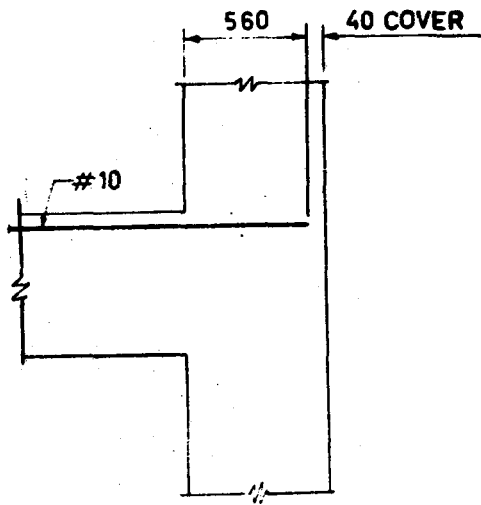
A few examples concerning development length in tension are illustrated in the following figures:



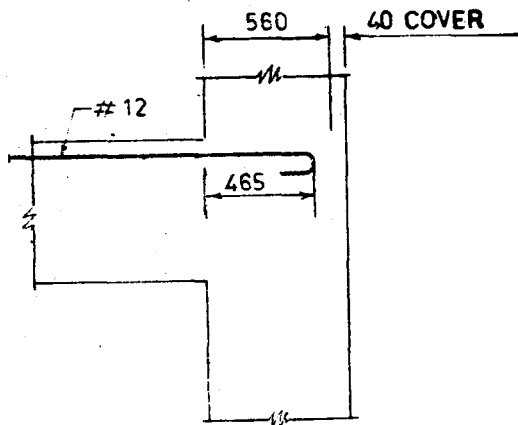
Available space for straight bar =
 $1150 - 25 = 1125$
 L_{dt} required (for # 20) = 1128
∴ Straight # 20 will fit



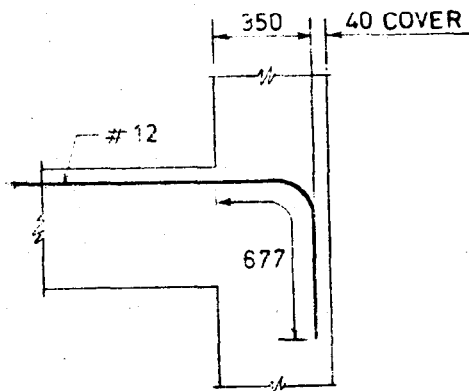
Available space for straight bar =
 $1400 - 25 = 1375$
 L_{dt} required (for # 28) = 1580
Straight # 28 will not fit
 $(L_{dt} - L_e) = 1580 - 224 = 1356$
∴ # 28 Bar with standard bend will fit



Available space for straight bar = 560
 L_{dt} required (for # 10) = 564
 $\therefore$ Straight bar # 10 will fit



Available space for straight bar = 560
 L_{dt} required (for # 12) = 677
 Straight bar will not fit
 $(L_{dt} - L_e) = (677 - 192) = 485$
 $\therefore$ # 12 with standard hook will easily fit



Available space for straight bar = 350
 L_{dt} required (for # 12) = 677
 $(L_{dt} - L_e) = (677 - 192) = 485$
 Straight bar with or without standard hook/bend
 will not be suitable
 $\therefore$ Provide # 12 bar with full embedded length =
 677 and also check for bearing stress at the end

(Page 41, clause 4.5.2) — Add the following at the end of the clause:

‘However this should be subject to the following conditions:

- Where the bar does not extend beyond a point four bar-diameters past the end of the bend; and
- Where the bar is assumed not to be stressed beyond a point four bar-diameters past the end of the bend at the ultimate design stress, that is, where the length of the bar extends beyond 4ϕ from end of the bend, it is not considered for development length.’

(Page 46, Fig. 4.18) — Delete ‘STANDARD’ in the legends.

(Page 47, Fig. 4.19) — Delete ‘STANDARD’ in the legends.

(Page 70, Fig. 6.1, Note 1) — Delete ‘standard’.

(Page 71, Fig. 6.2, Note) — Delete ‘standard’.

(Page 79, Fig. 6.11, SECTION AA) — Delete ‘STANDARD’ in the legend.

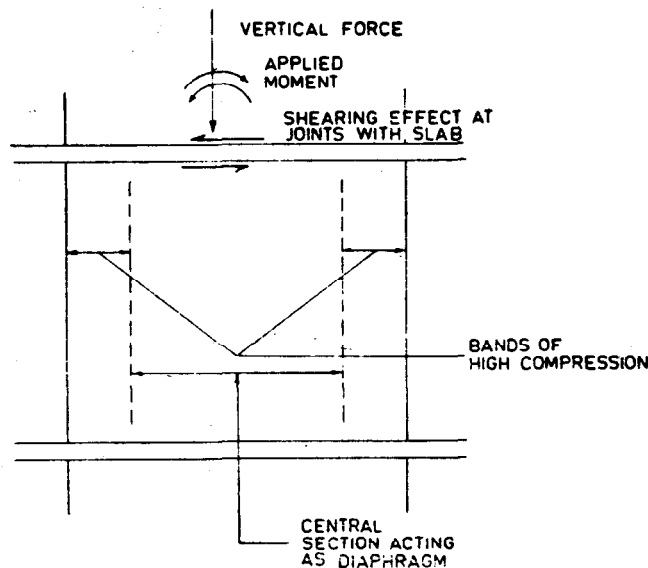
Free (Page 81, Fig. 6.13) — Substitute 'LAP SPLICES AT MID SPAN, IF REQUIRED' for 'LAP SPLICES AT MID SPAW, IF REQUIRED' and the same should be referred to top steel instead of bottom steel as shown by the arrow line.

(Page 129, Fig. 9.7A) — Delete 'STANDARD' in the legend.

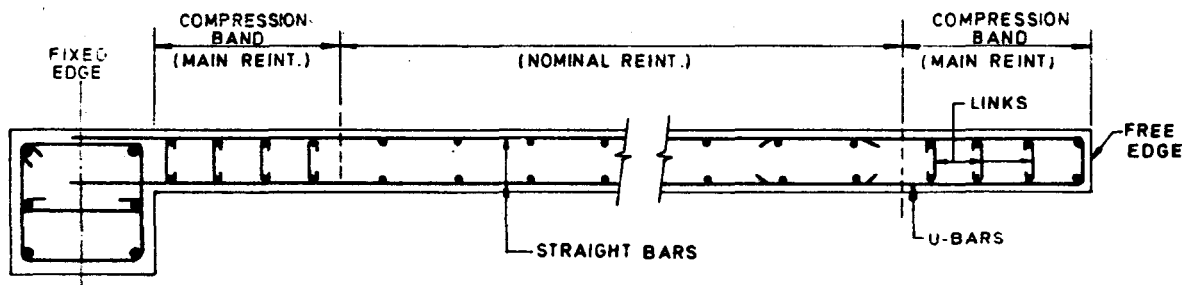
(Pages 181 to 185, Fig. 11.29 to 11.33) — Change Fig. 11.29 to Fig. 11.32; Fig. 11.30 to Fig. 11.29; Fig. 11.31 to Fig. 11.30; and Fig. 11.32 to Fig. 11.31.

(Page 181, clause 11.9, line 5) — Substitute ' Fig. 11.34 ' for ' Fig. 11.32 '

(Page 185) — Add the following Fig. 11.34 below Fig. 11.33:



11.34A Forces in Shear Wall



11.34B Cross Section Showing Typical Details
FIG. 11.34 SHEAR WALL

(Page 227, Table C-1, Sl No. 3) — Substitute '5'0' for '3'0' in col 3.

(Page 449, Sheet 8) — Substitute '(SEE SHEET 10 FOR REINFORCEMENT SCHEDULE)' for '(See SHEET 9 FOR REINFORCEMENT SCHEDULE)' in the title block.

(Page 253, Sheet 10) — Substitute '(REFER SHEET 8 FOR ARRANGEMENT OF REINFORCEMENT)' for '(REFER SHEET 9 FOR ARRANGEMENT OF REINFORCEMENT)'.